QA / TEST RECORD

AO-01 — Test results

Smart Alert Correlation & Root-Cause Engine · 29 October 2026

Live application: https://crash-monitor.vercel.app/ (HTTP 200 checked 29 Oct 2026; title: AO-01 | Incident Overview)

23 cases	**22** passed	**1** failed	**Backend suite** · 7 passed	**Frontend build** · succeeded

Finding to address: TC-017 — the webhook accepts an empty JSON object as one generic alert. Expected behavior is HTTP 400 with no ingestion. This is reported as observed; it has not been changed.

The existing backend pytest suite passed all 7 tests. API checks used FastAPI TestClient with the project dataset. The production frontend build succeeded with a large JavaScript chunk advisory.

Case / type	Input	Expected result	Actual result / status	Why this test matters
TC-001 Positive **Privacy — IPv4 redaction**	“Failed to connect to 192.168.1.105:5432”; redaction on.	Replace the IP with a pseudonym and retain the port.	**PASS** **IP became a stable ip- pseudonym; port remained.**	Keeps private network details out of alert text.
TC-002 Positive **Privacy — hostname redaction**	“srv-payments-db-42.prod.internal went down”; redaction on.	Replace the internal hostname with a pseudonym.	**PASS** **Hostname became a host- pseudonym.**	Prevents internal infrastructure names from appearing in output.
TC-003 Positive **Privacy — customer IDs**	Text containing tenant_enterprise99 and cust-492184.	Pseudonymize both identifiers.	**PASS** **Both were replaced with cust- pseudonyms.**	Protects tenant and customer identifiers in alert content.
TC-004 Boundary **Privacy — redaction off**	“Secret host db-master.internal with IP 10.0.0.1”; redaction off.	Return the original text unchanged.	**PASS** **Output matched the original.**	Checks behavior at the disabled setting.
TC-005 Validation **Ingestion — alert-name normalization**	DBQueryTimeout_1234_node-a8f9b2.	Remove transient IDs and host tokens; preserve the stable rule name.	**PASS** **Digits and host token removed; dbquerytimeout retained.**	Consistent names let repeated alerts map to one rule.
TC-006 Validation **Noise — rule-key construction**	Prometheus, Orders-DB, HighLatency_P99_4891.	Build the canonical lowercase rule key.	**PASS** **Returned prometheus:orders-db:highlatency p.**	Stable keys support grouping and review.
TC-007 Positive **Correlation — recall safety**	Run the synthetic ground-truth backtest with recall guard.	100% incident recall, zero misses, and over 50% notification reduction.	**PASS** **Automated acceptance assertions passed.**	Checks that real incidents stay visible while alert volume falls.
TC-008 Validation **Frontend — production build**	Run npm run build from the project root.	TypeScript compilation and Vite build complete.	**PASS** **Build completed; Vite printed a large-chunk advisory.**	Catches compile errors before deployment.

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TC-009 Positive **API — health**	GET /api/health.	HTTP 200 with service health details.	**PASS** **HTTP 200; healthy status and service metadata returned.**	Confirms the API starts and reports its state.
TC-010 Boundary **Incident feed — one-item page**	GET /api/incidents?limit=1&offset=0.	Return at most one incident.	**PASS** **Returned no more than one incident.**	Checks pagination at a one-record page size.
TC-011 Boundary **Incident feed — offset past end**	GET /api/incidents?limit=10&offset=999999.	Return HTTP 200 and an empty incident list.	**PASS** **HTTP 200; incidents array was empty.**	Out-of-range pages should be handled safely.
TC-012 Negative **Incident detail — unknown ID**	GET /api/incidents/not-a-real-id.	Return HTTP 404.	**PASS** **HTTP 404; “Incident not found”.**	Ensures missing records produce a clear response.
TC-013 Positive **Service map — topology**	GET /api/topology.	Return HTTP 200 with topology nodes and edges.	**PASS** **HTTP 200; service nodes were returned.**	Provides data needed to render the dependency map.
TC-014 Positive **Noise review — rules and what-if**	GET /api/noise/rules and GET /api/noise/what-if.	Both endpoints return HTTP 200.	**PASS** **Both returned HTTP 200.**	Checks the data behind recommendations and impact simulation.
TC-015 Positive **Performance — benchmark scoreboard**	GET /api/metrics.	Return HTTP 200 with four benchmark methods.	**PASS** **HTTP 200; four benchmark rows returned.**	Checks the scoreboard data contract.
TC-016 Negative **Webhook — malformed JSON**	POST /api/webhook/alert with body “not-json”.	Return HTTP 400 without ingesting an alert.	**PASS** **HTTP 400; invalid JSON response.**	Rejects malformed integration payloads.
TC-017 Validation **Webhook — empty object**	POST /api/webhook/alert with JSON {}.	Return HTTP 400; ingest no records.	**FAIL** **HTTP 200; empty object became one generic alert.**	Empty payloads must not create misleading synthetic alerts.
TC-018 Validation **Configuration — invalid type**	PUT /api/config with {"W":"bad"}.	Reject with HTTP 422; do not apply the value.	**PASS** **HTTP 422 validation response.**	Prevents invalid settings from being accepted.
TC-019 Boundary **Privacy — redaction toggle**	PUT /api/redaction false, then true.	Return each requested value and finish enabled.	**PASS** **Both returned HTTP 200; final state restored to true.**	Checks controlled changes to privacy behavior.
TC-020 Positive **Webhook — valid alert**	POST one valid alert record to /api/webhook/alert.	Return HTTP 200; ingest and correlate one alert.	**PASS** **HTTP 200; one alert ingested and correlated.**	Verifies the normal monitoring-integration path.

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TC-021 Negative **File upload — empty CSV**	Upload an empty file named empty.csv to /api/ingest/upload.	Return HTTP 400; load no alerts.	**PASS** **HTTP 400; no valid alert records found.**	Rejects files that contain no alert records.
TC-022 Positive **File upload — valid JSON**	Upload one alert with timestamp, service, severity, name, and message.	Return HTTP 200; load one alert and create an incident.	**PASS** **HTTP 200; one alert loaded and one incident created.**	Confirms supported files can be ingested and correlated.
TC-023 Validation **Noise review — decision audit**	POST decision rejected for a valid rule key.	Record the decision without changing live monitoring rules.	**PASS** **HTTP 200; audit-only execution note returned.**	Keeps review actions auditable without external rule changes.